

Video 14 – Support Vector Machines

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Video Synopsis

SVMs and the concept of the margin -11:20

The core concept in SVM's of a margin around the dividing line/plane is introduced and it is argued that a large margin is best as it results in the best chance that a noisy data point will still end up on the correct side of the line/plane that divides the two classes. The concept of growth functions is used to show that the requirement of a 'fat' margin results in fewer dichotomies being allowed, thus limiting the growth function. And we know that this means our VC dimension is limited, which is good for generalization.

SAFE SKIP! Solving for weights that result in dividing plane with biggest margin 11:20-50:05

SAFE SKIP The SVM learning algorithm is introduced as a mathematical optimisation problem in which the boundary is positioned such that it is as far away as possible of the nearest points in both classes. This leads to a constrained quadratic optimisation problem, which can be solved using Lagrange multipliers.

The optimisation generally leads to a solution (model) of lower complexity 50:05-52:53

Prof. Abu-Mustafa makes an interesting observation: the original decision function which contained all data points, and thus is potentially very complex, is reduced to a generally far smaller summation as most terms will be zero. Comparing this to the weights of e.g. logistic regression, we can understand that the result is beneficial from a complexity perspective.

Adding features without paying the complexity penalty.. 52:53-1:05:17

The concept of casting the data in a higher space is introduced to improve linear separability. Prof Abu Mostafa explains that this casting in a higher space (which effectively means adding extra features) comes at a very low price: as the SVM learning algorithm uses an inner product of the feature vector with itself, the result is always a scalar. So the only penalty is in having to perform an inner product with a longer vector and this is a very minor penalty to incur.

It is then shown that finding a separating hyperplane in this higher dimensional space, may result in a very wiggly decision boundary in the original space. The knee-jerk response would be to conclude that the model is heavily overfitted, but this is (likely) not the case: it is the number of support vectors used that defines the complexity of the model (and thus the potential for overfitting). An upper limit for the expected value of E_{out} is introduced as the number of support vectors divided by the number of training samples -1:

$$\mathbb{E}[E_{out}] \leq \mathbb{E}\left[\frac{\#of\ SVs}{N-1}\right]$$

Hence, the generalization obtained with the SVM can be checked after training is completed by looking at the support vectors found.

SAFE SKIP! Q&A

1:05:17 - end